

Generate a Lecture Scribe for exam reference from the given PPT L26 using the structure from the PDF "Example_Scribe". Follow t

Example_Scribe



L26



Generate a Lecture Scribe for exam reference from the given PPT L26 using the structure from the PDF "Example_Scribe". Follow the below two sets of instructions properly to generate the Lecture Scribe :

Instructions : 1) The lecture scribe is to be generated as per the structure given in Example_Scribe 2) Use the content from L26 PPT only and avoid content outside ppt

Instructions for following the example scribe : 1) For List of topics covered, the lecture scribe must cover the list of all topics covered in the PPT L26 Do not add new headings/topics that are not covered in ppt 2) For Explanation of Topics Covered, the lecture scribe must cover explanation of topics covered in PPT 3) For List of Definitions and Theorems, the lecture scribe must include detailed definitions with notations/assumptions along with exact statement of definitions and theorems, detailed proof of theorem step by step (proof only if given in ppt) 4) For Important Examples, try to cover more number of examples from the PPT attached (L26) in such a way that it covers all the concepts discussed. Write entire question for the example, as given in L26 PPT and write detailed step by step solution to examples. INCLUDE EXAMPLES FROM PPT ONLY 5) For the List of Important Formulas, write the list of all the important formulas with reference to topics covered in PPT (as per example_scribe format)

◆ Lecture Scribe: Stochastic Processes - SSS, WSS, and Ergodic Processes

1) List of Topics Covered

1. Basics of Stochastic Processes

- Concept of Waveform Collections and Realizations
- Process Characterization (First, Second, and Nth Order Density Functions)

2. Statistical Moments

- Mean Vector (Mean Function)
- Autocorrelation Function
- Autocovariance Function
- Correlation Coefficient
- Autocorrelation Properties and Proofs

3. Stationary Stochastic Process

- Strict Sense Stationary (SSS) Process
- Wide Sense Stationary (WSS) Process

4. Ergodic Process

- Time Average of a Random Process
 - Mean Ergodicity and Autocorrelation Ergodicity
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2) Explanation of Topics Covered

1. Basics of Stochastic Processes

A **Stochastic Process** $X(\zeta, t)$ is a collection of time functions (waveforms) associated with the outcomes of a random experiment.

- **Realization/Sample Function:** For a fixed outcome ζ_i , $X(\zeta_i, t)$ is a deterministic function of time.
- **Random Variable:** At a fixed time t_k , $X(\zeta, t_k)$ is a random variable representing the values across all realizations at that specific instant.

2. Statistical Moments

To characterize a process without knowing the full Nth order PDF, we use moments:

- **Mean Function:** The expected value of the process at any time t . It represents the ensemble average across all possible waveforms.
- **Autocorrelation:** Measures the interdependence of the values of the process at two different times, t_1 and t_2 .
- **Autocovariance:** Measures the correlation of the fluctuations of the process around its mean at two different times.

3. Stationary Stochastic Process

A process is **Stationary** if its statistical properties do not change over time (time-invariance).

- **Strict Sense Stationary (SSS):** A process where the joint distribution of any set of samples is invariant to a shift in the time origin.
- **Wide Sense Stationary (WSS):** A less restrictive form where only the first and second moments (mean and autocorrelation) are time-invariant.

4. Ergodic Process

A process is **Ergodic** if its time averages (calculated from a single realization over a long period) are equal to its ensemble averages (calculated across all realizations at a specific time). This is crucial in practice because we often only have access to one long signal trace.

3) List of Definitions and Theorems

Definition: Mean Function

The mean of a random process $X(t)$ is the expectation of the random variable $X(t)$ for a fixed t :

$$\mu_X(t) = E[X(t)] = \int_{-\infty}^{\infty} x f_{X(t)}(x) dx$$

Definition: Autocorrelation Function

The autocorrelation $R_X(t_1, t_2)$ of a random process $X(t)$ is the joint moment of the random variables $X(t_1)$ and $X(t_2)$:

$$R_X(t_1, t_2) = E[X(t_1)X(t_2)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x_1 x_2 f_{X(t_1), X(t_2)}(x_1, x_2) dx_1 dx_2$$

Definition: Autocovariance Function

$$C_X(t_1, t_2) = E[(X(t_1) - \mu_X(t_1))(X(t_2) - \mu_X(t_2))] = R_X(t_1, t_2) - \mu_X(t_1)\mu_X(t_2)$$

Definition: Wide Sense Stationarity (WSS)

A process $X(t)$ is WSS if:

1. The mean is a constant: $E[X(t)] = \mu_X$
2. The autocorrelation depends only on the time difference $\tau = t_1 - t_2$:

$$E[X(t)X(t + \tau)] = R_X(\tau)$$

Theorem: Properties of Autocorrelation (for WSS)

1. **Symmetry:** $R_X(\tau) = R_X(-\tau)$
2. **Maximum Value at Origin:** $|R_X(\tau)| \leq R_X(0)$
3. **Mean Square Value:** $R_X(0) = E[X^2(t)]$

Proof of Symmetry:

$$R_X(\tau) = E[X(t)X(t + \tau)]$$

Let $t + \tau = t'$, then $t = t' - \tau$

$$R_X(\tau) = E[X(t' - \tau)X(t')] = E[X(t')X(t' - \tau)] = R_X(-\tau)$$

4) Important Examples

Example 1: Sinusoid with Random Phase

Question: Let $X(t) = A \cos(\omega_c t + \Theta)$, where A and ω_c are constants and Θ is a random variable uniformly distributed over $[0, 2\pi]$. Determine if the process is WSS.

Solution:

1. **Check Mean:**

$$\mu_X(t) = E[A \cos(\omega_c t + \Theta)] = \int_0^{2\pi} A \cos(\omega_c t + \theta) \frac{1}{2\pi} d\theta$$

$$\mu_X(t) = \frac{A}{2\pi} [\sin(\omega_c t + \theta)]_0^{2\pi} = \frac{A}{2\pi} [\sin(\omega_c t + 2\pi) - \sin(\omega_c t)] = 0$$

Since the mean is constant (0), the first condition for WSS is met.

2. Check Autocorrelation:

$$\begin{aligned}R_X(t, t + \tau) &= E[X(t)X(t + \tau)] \\&= E[A \cos(\omega_c t + \Theta) \cdot A \cos(\omega_c(t + \tau) + \Theta)] \\&= \frac{A^2}{2} E[\cos(\omega_c \tau) + \cos(2\omega_c t + \omega_c \tau + 2\Theta)]\end{aligned}$$

Using linearity of expectation:

$$= \frac{A^2}{2} \cos(\omega_c \tau) + \frac{A^2}{2} E[\cos(2\omega_c t + \omega_c \tau + 2\Theta)]$$

The second term involves an integral over two periods of a cosine, which is 0.

$$R_X(\tau) = \frac{A^2}{2} \cos(\omega_c \tau)$$

Since R_X depends only on τ , the process is **WSS**.

Example 2: Time Average

Question: Given a realization $x(t) = A \cos(\omega_c t + \theta)$, find the time-averaged mean.

Solution:

$$\begin{aligned}A[x(t)] &= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T A \cos(\omega_c t + \theta) dt \\A[x(t)] &= \lim_{T \rightarrow \infty} \frac{A}{2T\omega_c} [\sin(\omega_c t + \theta)]_{-T}^T = 0\end{aligned}$$

Note: Since the ensemble mean $E[X(t)] = 0$ and the time average $A[x(t)] = 0$, the process is **mean-ergodic**.

5) List of Important Formulas

Topic	Formula
Mean Function	$\mu_X(t) = E[X(t)]$
Autocorrelation	$R_X(t_1, t_2) = E[X(t_1)X(t_2)]$
Autocovariance	$C_X(t_1, t_2) = R_X(t_1, t_2) - \mu_X(t_1)\mu_X(t_2)$
Correlation Coefficient	$\rho_X(t_1, t_2) = \frac{C_X(t_1, t_2)}{\sigma_X(t_1)\sigma_X(t_2)}$
WSS Mean	$E[X(t)] = \text{constant}$
WSS Autocorrelation	$R_X(\tau) = E[X(t)X(t + \tau)]$

Topic	Formula
Time Average (Mean)	$\bar{x} = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T x(t) dt$